Article

Matter, Solitons, and Emergent Mass in a Deterministic Brane-Lattice Substrate (Paper IV)

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Abstract

We develop the matter and mass consistency bridges for the deterministic 4D brane-lattice substrate of Paper I. The matter bridge poses particle-like states as self-adjoint eigenproblems with angular structure organized by vector spherical harmonics (VSH). Two mechanisms together establish Derrick stability: anti-collapse is provided by the lattice spacing a (no continuous Derrick rescaling below one cell), and anti-expansion by the geometric quartic $\propto k_s\alpha/a$, giving an equilibrium soliton radius $R_h/a \approx \kappa(A/a)\sqrt{\alpha/(1-\alpha)}$ with a soliton sweet spot at $\alpha \approx 0.5$–0.8. The canonical baryon ansatz is the hedgehog $\xi^i = f(r)\hat{x}^i$ ($J = 0, L = 1$ VSH), which lives entirely in the $SU(3)$ traceless sector and carries neutron-like far-field holonomy; a proton-like far field requires a nonzero $L = 0$ scalar admixture. Soliton states carry labels $(J, P;\ SU(3)\text{-irrep};\ Q_{U(1)};\ B_{\text{winding}})$ with charge-triality locking $q \equiv -t \pmod 3$ inherited from Paper III. Periodic boundary conditions are mandatory: open BC allows the prestressed lattice to relax (contracting to αa), collapsing even the $A = 0$ vacuum. The mass bridge derives rest mass from self-confinement, not from the Higgs mechanism: a wave permanently trapped in a closed-loop toroidal mode has a total substrate energy $E(R^*, A) = mc^2$ at the equilibrium major radius $R^* = \sqrt{E_{\text{curv}}/(2\pi T_{\text{string}})}$. The complex structure $\Psi \approx \xi + (\mathrm{i}/\omega_0)\dot{\xi}$ originates entirely in the timelike worldtube direction; the $U(1)$ carrier frequency ω_0 is fixed by the worldtube closure condition. Time-link prestress sources a binding compression $\Delta u_{\parallel}^{\text{ext}} \propto -\alpha\omega^2 < 0$ that flips the $U(1)$–$SU(3)$ coupling to attractive when $\chi = 2\alpha(\omega/\omega_*)^2(A_0/a)^2 > 1$. Both bridges are gated on the same object: a converged stable baryon-like eigenstate, the decisive numerical target of the series.

Keywords: soliton; Derrick stability; vector spherical harmonics; hedgehog ansatz; Skyrme texture; emergent mass; self-confinement; toroidal mode; time-link binding; brane lattice

1. Introduction

This is the fourth and final paper in the series. Paper I established the substrate model. Papers II and III developed the Lorentz, gravity, gauge, and color bridges. The present paper develops two bridges that have been deliberately held back: matter and mass.

Matter bridge. The matter bridge poses particle-like states as self-adjoint eigenproblems on a symmetric background, with angular structure organized by vector spherical harmonics (Section 3). The key structural inputs are the geometric quartic $W_4 \propto k_s\alpha/a$ from Paper I and the $U(3)$ triplet structure from Paper III. Two mechanisms together establish stability: (i) anti-collapse is provided by the lattice spacing a — the Derrick $\lambda \to 0$ rescaling halts at one cell, because configurations concentrated below one cell are not local energy minima of the spring network; (ii) anti-expansion is provided by the geometric quartic $\propto k_s\alpha/a$, which

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under Derrick rescaling scales as λ^{-1} and overpowers the λ^{+1} gradient term for $\lambda > \lambda^*$. The equilibrium soliton radius from this balance is $R_h/a \approx \kappa\,(A/a)\sqrt{\alpha/(1-\alpha)}$, setting a soliton sweet spot at $\alpha \approx 0.5$–0.8.

Mass bridge. The mass bridge derives emergent rest mass from self-confinement of a closed-loop wave mode — not from the Higgs mechanism (Section 4). A wave trapped in a circularly closed toroidal excitation cannot disperse: it is pinned by topology and geometric self-guidance. The total substrate energy of this mode at equilibrium, $E(R^*, A)$, is identified with the rest mass mc^2. The complex structure $\Psi \approx \xi + (i/\omega_0)\dot{\xi}$ that mediates this picture arises entirely from the timelike worldtube direction — the imaginary unit i here is a quarter-period time shift, not an independent complex field. A separate, quantitative open problem is connecting the equilibrium radius R^* to the Compton wavelength $\hbar/mc$; this requires deriving $\hbar$ and c from substrate parameters consistently.

Why matter and mass are paired. Both bridges are gated on the same object: a converged, stable, localized soliton eigenstate. Without such an eigenstate, the mass identification $E(R^*, A) = mc^2$ remains schematic. The matter bridge's decisive numerical task (does the substrate support a stable baryon-like π_3 texture?) is therefore simultaneously the gate that unlocks the mass bridge's normalization argument. Pairing them in one paper makes this dependency explicit.

Paper organization. Section 2 lists imports from Papers I and III. Section 3 develops the matter bridge: self-adjoint eigenproblem, VSH angular structure, Derrick stability, soliton labels, and periodic-BC requirement. Section 4 develops the mass bridge: self-confinement picture, toroidal mode energy, Derrick balance for torus geometry, time-link binding, and open normalization problems.

2. Imported interfaces

This paper imports the following results from Papers I and III without re-derivation.

From Paper I

Equations of motion. Equation (26).

Geometric nonlinearity and quartic coefficient. The constitutive quartic $W_4 \propto k_s\alpha/a$ (Equation (20)), which provides the anti-expansion stabilization of any localized mode.

Induced metric and near-identity expansion. Equations (10) and (29); the transverse–lateral coupling $\partial_i u\,\partial_j u$ underlies both the gravity channel (Paper II) and the geometric self-guidance discussed here.

Continuum action. Equation (24).

Discrete lattice action. Equation (7); the lattice spacing a is the UV cutoff that blocks Derrick collapse.

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From Paper III

$U(3)$ triplet decomposition. The axis-aligned lateral triplet $\Psi \in \mathbb{C}^3$, its $\mathfrak{u}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3)$ decomposition, and the $U(1)$ trace identification with electromagnetism (Paper III Equations (12) and (13)).

Kinematic color confinement. The result that color charge cannot be carried to infinity by a free linear excitation (Paper III Section 4.4).

Charge-triality locking. $q \equiv -t \pmod 3$, which constrains the soliton labels defined in Section 3.4.

Gauge-covariant envelope dynamics. The covariant derivative $\mathcal{D}_\mu \Psi = (\partial_\mu + \mathrm{i}\mathcal{A}_\mu)\Psi$ and effective action S_{eff} (Paper III Equations (10) and (11)).

3. Matter bridge: solitons as self-adjoint eigenproblems

3.1. Linearization about symmetric backgrounds

Let $\mathbf{X}_0(x)$ be a time-independent background embedding representing a localized deformation region, taken to be spherically symmetric in the intrinsic brane coordinates ($r = |x|$ with $r \gg a$ for the continuum limit to apply). Linearize the equations of motion (Equation (26) of Paper I) about $\mathbf{X}_0$ by writing $\mathbf{X}(x,t) = \mathbf{X}_0(x) + \boldsymbol{\eta}(x,t)$ with small perturbation $\boldsymbol{\eta}$. Because the dynamics derives from the action (Equation (24) of Paper I), the linearized operator is the second variation of S and is therefore self-adjoint with respect to the ρ_m-weighted L^2 inner product on Ω; orthogonality and completeness of eigenmodes follow as standard consequences.

3.2. Angular structure: vector spherical harmonics

The substrate is the 6-neighbor cubic lattice, whose point group is O_h rather than $SO(3)$. At scales $\gg a$, however, long-wavelength continuum modes recover approximate isotropy with residual cubic corrections of order 7.5% in longitudinal speed at $\alpha = 0.2$. Solitons in this theory are expected to live in the geometric-quartic-stabilized regime with $\ell_{\mathrm{soliton}} \gg a$, so the emergent approximate $SO(3)$ is the right symmetry to organize their angular structure, with O_h corrections entering as small splittings of $SO(3)$ multiplets.

For a spherically symmetric background the linearized operator commutes with the action of $SO(3)$ on the intrinsic coordinates at leading order in $a/\ell_{\mathrm{soliton}}$. For a scalar field this gives the familiar separation into radial and angular problems with eigenfunctions $Y_{\ell m}(\theta, \varphi)$. The lateral triplet $\xi^i \in \mathbb{R}^3$ and its complex-envelope promotion $\Psi \in \mathbb{C}^3$ from Paper III carry an internal vector index i in addition to spatial dependence. The correct angular basis is therefore the *vector spherical harmonic* (VSH) basis $\mathbf{Y}^i_{JLM}$, where J is total angular momentum, L is orbital angular momentum, and $L \in \{J-1, J, J+1\}$:

$$\xi^i(r,\theta,\varphi) \;=\; \sum_{J,L,M} f_{JLM}(r)\,\mathbf{Y}^i_{JLM}(\theta,\varphi). \tag{1}$$

The choice of (J, L) specifies how the internal index i is glued to the spatial angular coordinates — it is the choice of how the triplet's $U(3)$ gauge structure (Paper III) couples to the geometry of the localized mode. Configurations with $J = 0$ are spherically symmetric only under the *combined* rotation that acts on both spatial and internal indices together; this combined rotation is the physically meaningful notion of spherical symmetry for a triplet field.

3.3. Hedgehog ansatz and Skyrme-twisted extension

The canonical baryon ansatz is the *hedgehog*

$$\xi^i(\mathbf{x}) \;=\; f(r)\,\hat{x}^i, \tag{2}$$

with $J = 0$, $L = 1$: the internal index i is locked to the spatial direction $\hat{x}^i$, so the configuration is invariant only under the combined diagonal $SO(3)$. The entire nontrivial structure of the hedgehog lives in the $SU(3)$ traceless sector of Paper III: the $U(1)$ trace $f(r)\,(\hat{x}^1 + \hat{x}^2 + \hat{x}^3)$ averages to zero on the sphere, giving a trace-neutral far field (neutron-like far-field holonomy).

Topological stabilization at winding number $B = 1$ is achieved by extending the hedgehog into a *Skyrme-twisted* configuration in which the embedding-amplitude direction X^4 also rotates radially:

$$\big(\xi^i, u\big)(\mathbf{x}) \;=\; \big(f(r)\sin F(r)\,\hat{x}^i,\; f(r)\cos F(r)\big), \tag{3}$$

with the profile $F : [0, \infty) \to [\pi, 0]$ enforcing the boundary conditions $F(0) = \pi$ and $F(\infty) = 0$. This boundary condition gives $B = 1$ topological winding in $\pi_3(SU(3)) = \mathbb{Z}$, consistent with Paper III Section 4.3. Adding a nonzero $L = 0$ scalar admixture shifts the $U(1)$ far-field holonomy from trace-neutral to proton-like.

3.4. Soliton labels

A localized 3D mode is fully classified at the soliton layer by the labels

$$\big(J,\, P;\; \mathrm{SU}(3)\text{-irrep};\; Q_{U(1)};\; B_{\text{winding}}\big), \tag{4}$$

where:

- J is total angular momentum from the emergent approximate $SO(3)$ at scales $\gg a$; it refines to an O_h irrep at sub-leading order in a/ℓ_{soliton}.
- P is parity, the discrete element of the emergent rotation group.
- The $SU(3)$ irrep and $U(1)$ charge come from the $U(3)$ decomposition (Paper III Section 4.2).
- B_{winding} is the topological winding number in $\pi_3(SU(3)) = \mathbb{Z}$.

These labels are not independent: the VSH coupling sets a Clebsch–Gordan relationship between $SO(3)$ and $U(3)$ representations, and the charge-triality locking $q \equiv -t \pmod 3$ (Paper III Section 4.2) constrains $Q_{U(1)}$ given the $SU(3)$ irrep.

3.5. Stability: Derrick's theorem on the discrete substrate

In 3D, a standard Derrick scaling argument [1] rules out stable finitely-extended solitons in a purely quadratic (linear) theory: under $x \mapsto \lambda x$, the gradient energy E_2 scales as λ^1 (favoring expansion for $\lambda > 1$) while a quartic energy E_4 scales as λ^{-1} (resisting expansion). Without $E_4 > 0$, the soliton expands without bound. The present substrate breaks the standard Derrick counting in two complementary ways.

Anti-collapse: lattice UV cutoff. Derrick's argument assumes continuous rescaling $\lambda \in (0, \infty)$. On a discrete lattice with spacing a, however, λ is bounded below: a soliton cannot shrink past one cell. A configuration with energy fully concentrated at a single node is not a local energy minimum — the spring network immediately disperses it outward. The lattice spacing a is the physical UV cutoff; no separate quartic repulsion is needed to block collapse. This is a genuinely discrete-substrate effect that has no direct continuum analogue.

Anti-expansion: geometric quartic. Under Derrick rescaling in 3D, the geometric quartic $W_4 \propto k_s\alpha/a$ (Paper I Equation (20)) scales as λ^{-1}. Balancing λE_2 against $\lambda^{-1}E_4$ gives equilibrium at

$$\lambda^* = \sqrt{E_4/E_2}, \tag{5}$$

translating to an equilibrium soliton half-radius

$$\frac{R_h}{a} \;\approx\; \kappa\,\frac{A}{a}\sqrt{\frac{\alpha}{1-\alpha}}, \tag{6}$$

where κ is a numerical factor of order unity fixed by the Derrick balance in the specific ansatz and A is the peak amplitude. At $\alpha = 0.2$ and $A/a \approx 10$ this gives $R_h/a \approx 5$; the soliton sweet spot $\alpha \approx 0.5$–0.8 gives $R_h/a \approx 8$–14 for the same amplitude, well above the lattice scale.

Note that the quartic coefficient scales as $\propto \alpha/a$, not $\propto (1-\alpha)$ as a naive StVK identification would give. The correct sign and scaling are derived from the spring's geometric quartic in Paper I (Equation (20)), which inverts the naïve StVK prediction. Quantitative soliton sizing must use the spring law, not StVK.

3.6. Periodic boundary conditions: mandatory requirement

Periodic boundary conditions (BC) are mandatory for all nonlinear soliton simulations. The prestressed lattice (rest length $\alpha a < a$) is under tension; open/free BC allows the box to contract from Na to $N\alpha a$, which dominates all dynamics and collapses even the $A = 0$ vacuum. Periodic BC fixes the box at $N \cdot a$, holds the tension, and ensures the vacuum is stable ($A = 0 \Rightarrow E_{\text{excess}} \equiv 0$). All retracted 2026-06-05 runs used open BC; all future nonlinear runs must use periodic-clamped BC.

3.7. Matter bridge open problems

1. **Converged baryon eigenstate.** Obtain a converged, stable, localized π_3-winding solution using the periodic-clamped tensioned vacuum and an eigenstate approach (not seed-and-watch evolution). The hedgehog/Skyrme-twisted ansatz (3) with $\alpha \approx 0.7$, $A/a \approx 10$ is the primary candidate.
2. **Rigorous Derrick stability proof.** State Derrick's theorem precisely [1,2] for the present action, give the lattice-UV anti-collapse argument as a theorem (not just intuition), and tie the anti-expansion balance (5) to the effective field manifold of Paper III.
3. **Converged $U(1)$ vortex eigenstate.** Resolve the semilocal drift observed in the preliminary JFNK run (Paper III Section 4.5) with branch-selection machinery for the semilocal vortex.
4. **Two-soliton interaction potential.** Once individual stable eigenstates exist, measure the two-soliton interaction potential as a function of separation and polarization alignment, to test whether the geometric self-guidance mechanism produces an attractive or repulsive channel.

4. Mass bridge: rest mass from self-confinement

4.1. Physical picture: wave trapped in a closed loop

Rest mass in this framework is not acquired through the Higgs mechanism (spontaneous symmetry breaking and a scalar condensate). Instead, rest mass is identified with the total substrate energy of a wave mode that is permanently trapped in a circularly closed, self-confined loop:

$$mc^2 = E[\mathbf{X}_{\text{torus}}]\Big|_{R=R^*}, \tag{7}$$

where $\mathbf{X}_{\text{torus}}$ is the toroidal mode embedding and R^* is the equilibrium major radius. The wave cannot disperse because it is topologically pinned (the $U(1)$ vortex carries $\pi_1(U(1)) = \mathbb{Z}$ winding) and geometrically self-confined (the geometric quartic prevents radial expansion). This is closer in spirit to the toroidal-electron proposal of Williamson and van der Mark [3], which attributes the correct gyromagnetic ratio $g = 2$ to a circularly self-confined photon-like mode, than to the standard Higgs picture. The substrate grounds that proposal in a concrete elastic medium.

4.2. Complex structure from the time direction

The complex structure $\Psi \in \mathbb{C}^3$ used throughout this series is not an independent complex field. It arises from the carrier rotation of a real substrate field along the timelike worldtube axis. For a band-isolated carrier at frequency ω_0, the complex envelope is defined by

$$\Psi \;\approx\; \xi + \frac{\mathrm{i}}{\omega_0}\,\dot{\xi}, \tag{8}$$

where ξ is the real lateral triplet and $\dot{\xi}$ is its time derivative. The imaginary unit i here is the quarter-period time shift (the time-evolution generator $\mathrm{i}\partial_t\Psi = H_{\mathrm{eff}}\Psi$); it is not an independent degree of freedom. A single spacelike slice is a real snapshot and carries no phase; the $U(1)$ phase exists only across the time extent of the worldtube. Two adjacent time slices are the minimum data that fix the $U(1)$ — exactly the structure of the rotating-frame-periodic BVP solver that must be used for toroidal mode eigenstates.

This origin of the complex structure has a direct consequence for mass: the $U(1)$ carrier frequency ω_0 is the rate of rotation of the real field along the time axis, and the energy stored in one carrier cycle is the substrate analogue of a quantum of energy. The mass-frequency relation $E \approx \hbar\omega_0$ (in appropriate units) is therefore expected to hold at the equilibrium amplitude, with $\hbar$ emerging from the substrate normalization (Section 4.5).

4.3. Toroidal mode energy and Derrick balance

A toroidal mode is a $U(1)$ vortex (line vortex in the trace sector) wrapped into a closed loop of major radius R and core radius w. The total substrate energy decomposes into gradient and quartic contributions:

$$E(R,A) \;=\; E_2(R,A) + E_4(R,A), \tag{9}$$

where at leading order in w/R:

$$E_2(R,A) \;\sim\; 2\pi R\cdot\varepsilon_2(A,w), \tag{10}$$

$$E_4(R,A) \;\sim\; 2\pi R\cdot\varepsilon_4(A,w), \tag{11}$$

with ε_2 and ε_4 the gradient and quartic energy densities per unit length of the vortex core. Both scale linearly with R (circumference grows), so to leading order the total energy is linear in R — this is the string-tension picture. The correction that selects a finite equilibrium R^* comes from the curvature energy of the torus: turning the line vortex into a closed loop costs an extra curvature term $\propto 1/R$, balancing the string-tension term $\propto R$:

$$E(R) \;\approx\; T_{\mathrm{string}}\cdot 2\pi R + \frac{E_{\mathrm{curv}}}{R}, \tag{12}$$

with T_{string} the vortex string tension set by $\varepsilon_2 + \varepsilon_4$ and E_{curv} the curvature energy from bending the vortex line. Setting $\mathrm{d}E/\mathrm{d}R = 0$ gives the equilibrium radius

$$R^* \;=\; \sqrt{\frac{E_{\mathrm{curv}}}{2\pi T_{\mathrm{string}}}}. \tag{13}$$

The physical rest mass is then $mc^2 = E(R^*) = 2\cdot 2\pi R^* T_{\mathrm{string}}$: the gradient and curvature contributions are equal at equilibrium.

4.4. Time-link prestress as binding agent

The timelike link in the discrete 4D brane action (Equation (7) of Paper I) carries a prestress $r_t = \alpha\beta\Delta t$ analogous to the spatial prestress parameter α. Expanding the time-link norm term $-k_t r_t|\Delta R_t|$ produces a negative, lump-co-located energy density in the carrier velocity field:

$$\bar{E}_{t,\perp}(\rho) \;=\; -\frac{m\alpha}{4}\,(\omega A(\rho))^2 \;<\; 0, \tag{14}$$

where ω is the carrier frequency fixed by the worldtube closure condition $\omega T = 2\pi n$ and $A(\rho)$ is the amplitude profile. This induces a definite-negative longitudinal compression on the spatial links:

$$\Delta u_{\parallel}^{\text{ext}}(\rho) \;=\; -\frac{m\alpha}{4k_s b^2}\,(\omega A(\rho))^2 \;\propto\; -\alpha\omega^2, \tag{15}$$

peaked at the vortex core $\rho \approx w$, zero on the axis. This is externally sourced by the time-link prestress and carrier eigenvalue, not slaved to the spatial relaxation.

Feeding (15) into the separable spatial cubic $V_3 = +(k_s\alpha/2a^2)\Delta u_{\parallel}(\xi_s^2 + |\xi_\perp|^2)$ flips the net cross-coupling:

$$g_{\text{net}} \;=\; \frac{k_s\alpha}{4a^2}\,(1-\chi), \qquad \chi \;=\; 2\alpha\,\frac{\omega^2}{\omega_*^2}\,\frac{A_0^2}{a^2}, \qquad \omega_* \equiv \frac{c_L}{b}. \tag{16}$$

Binding ($g_{\text{net}} < 0$, restoring force between $U(1)$ and $SU(3)$ sectors) holds iff $\chi > 1$. This is reachable at the soliton sweet spot $\alpha \approx 0.5$–0.8, ω near the band edge, $A_0/a \approx 0.3$. The spatial-only verdict (net repulsive) is the $r_t \to 0$ limit; the time link genuinely flips the sign.

Linchpin: closure-locked carrier frequency. The binding mechanism requires that ω be fixed externally by the worldtube closure condition $\omega T = 2\pi n$, not by spatial relaxation. This is the case for the rotating-frame-periodic BVP solver (`PeriodicBC/solve_block`): the JFNK root-find relaxes node positions only; ω is not a degree of freedom. Running the vortex through the breather eigensolver (where ω co-relaxes as a nonlinear eigenvalue) would violate this condition and revert the binding verdict to net-repulsive.

Binding-gravity co-variation. The same r_t (with $\alpha_t = \alpha$) sourcing the binding compression (15) also sources gravitational time dilation (Paper II): both coupling strengths scale as $\propto \alpha$. As a result, binding strength and gravitational time-dilation strength co-vary in α and are not independently tunable. If a simulation can tune binding without tuning gravity, the identification $\alpha_t = \alpha$ is falsified.

4.5. Open normalization problem: Compton wavelength

The mass-energy identification (7) requires three open normalization steps that are not completed here:

1. **Explicit toroidal energy integral.** Compute E_2, E_4, and E_{curv} in (12) explicitly by integrating the substrate energy density over the torus geometry with the hedgehog-like core profile of Section 3.
2. **Equilibrium amplitude.** The amplitude A is not fixed by the Derrick balance alone; it is selected by the energy normalization $E(R^*, A) = mc^2$ for a chosen target particle species. Express A in terms of substrate parameters $(k_s, \alpha, a, m_{\text{node}})$ and the target species mass.
3. **Compton wavelength connection.** Show that $R^* = \hbar/mc$ using consistent derivations of $\hbar$ and c from substrate parameters. Because $c^2 = k_s a^2/m_{\text{node}}$ and the natural action quantum is set by $k_s a^3$ (energy times volume) or $k_s a^2 \cdot (a/c)$ (energy times time), the combination giving $\hbar$ in SI units must close the triangle $R^* \cdot mc = \hbar$ without additional free parameters.

4.6. Mass bridge open problems

1. **Explicit toroidal energy and equilibrium radius.** Complete the calculation in Section 4.3 for the substrate spring law (Equation (20) of Paper I). Identify T_{string} and E_{curv} in terms of $(k_s, \alpha, a, w, \omega_0)$.
2. **Verify binding via closure-locked ω sweep.** Solve a stationary worldtube at each closure index n (each imposed $\omega = 2\pi n/(P\Delta t)$); on each converged state measure $\Delta u_{\parallel}(\rho)$ and the relaxed amplitude A. Confirm $\Delta u_{\parallel} \propto \omega^2$ across the converged family (dividing out A, which self-selects to the imposed ω).
3. **Winding-lock test.** Vary the topological winding B at fixed amplitude on a converged baryon texture; check whether the P-odd holonomy $\gamma_\Gamma \propto B$ (soft winding-lock) or is flat in B (energetic-only). A B-slope of 1 confirms charge tracks winding; flat-in-B indicates the energetic leg only.
4. **Compton wavelength derivation.** Carry out the three-step normalization of Section 4.5 and test whether $R^* = \hbar/mc$ closes without additional free parameters.

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Appendix A. Symbol dictionary

$\Omega \subset \mathbb{R}^3$	Intrinsic/material domain of the brane (coordinates $x = (x^1, x^2, x^3)$).
$\mathbf{X}(x,t) \in \mathbb{R}^4$	Embedding field of the brane into ambient 4D Euclidean space.
t	Slicing parameter along the inside observer's timelike direction ($t := x^4$); not an external/ambient time (the ambient is symmetric 4D Euclidean, Section 3.1).
g_{ij}	Induced metric: $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$.
g^0_{ij}	Isotropic reference metric: $g^0_{ij} = \ell_0^2\,\delta_{ij}$ (stress-free scale ℓ_0).
$h_\star$	Held ground-state scale (e.g. imposed by boundary conditions).
α	Dimensionless pre-stress parameter: $\alpha = \ell_0/h_\star \in (0,1]$.
$\widehat{g}$	Mixed relative metric: $\widehat{g} = (g^0)^{-1} g$.
E	Green–Lagrange strain: $E = \frac{1}{2}(\widehat{g} - I)$.
λ, μ	Effective Lamé parameters derived from the spring constitutive law: $\mu(\alpha) = (1-\alpha)k_s/a$, $\lambda = 0$ on the 6-neighbor stencil. StVK agrees at quadratic order only; see Section 3.5.
ρ_m	Brane mass density in the Lagrangian.
$W(g; g^0)$	Isotropic stored-energy density (hyperelastic).
$P_A^{\ i}$	First Piola stress: $P_A^{\ i} = \partial W / \partial(\partial_i X^A)$.
$\mathbf{u}$	Displacement field relative to a background: $\mathbf{u} = \mathbf{X} - \mathbf{X}_\star$.
u	Shorthand for a transverse component in a graph-type embedding, $\mathbf{X} \approx (x, u)$.
$\boldsymbol{\epsilon}$	Normalized polarization state of a narrowband multi-component wave.
a_μ	Berry (rank-1) connection: $a_\mu = \mathrm{i}\langle u \,\vert\, \partial_\mu u \rangle$.
$f_{\mu\nu}$	Berry curvature: $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$.
$\mathcal{A}_\mu$	Wilczek–Zee connection on an n-dimensional polarization subspace.
$\mathcal{F}_{\mu\nu}$	Wilczek–Zee curvature: $\partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - \mathrm{i}[\mathcal{A}_\mu, \mathcal{A}_\nu]$.
γ_Γ	Geometric phase / holonomy around loop Γ: $\gamma_\Gamma = \oint_\Gamma a_\mu\, \mathrm{d}x^\mu$.
Ω_{mix}	Mode-mixing (non-adiabatic conversion) rate between nearby polarization bands.
$\Delta\omega_{\mathrm{gap}}$	Local spectral gap to the nearest competing polarization band.
a	Lattice spacing of the cubic substrate (microphysical length scale).
$\tilde{a}$	Momentum-lattice spacing in the discretized Brillouin zone (typically $\tilde{a} \sim 2\pi/(N_s a)$).
$\mathbf{k}$	Crystal momentum / Brillouin-zone coordinate for lattice eigenmodes.
$\Psi = (\psi_x, \psi_y, \psi_z)^\top$	Axis-aligned complex narrowband triplet carrier (cubic lattice internal sector).
T^a	Generators of $\mathfrak{su}(3)$ (e.g. Gell–Mann basis) used to decompose a $U(3)$ connection.
$R^l_p \in \mathbb{R}^4$	4D node position in the explicit lattice: $p = (i,j,k)$ spacelike triple, $l \in \mathbb{Z}$ timelike index.
k_s	Central-force spring stiffness of the 6-neighbor spacelike stencil.
m	Node mass (discrete lattice; continuum mass density $\rho_m = m/a^3$).
Δt	Temporal lattice step (timelike link length in the explicit 4D lattice).
$\mathcal{N}_s$	6-neighbor axial spacelike stencil: $\{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$.
$\mathcal{N}_t$	Temporal neighbor stencil: $\{l \pm 1\}$.
c_L	Long-wavelength longitudinal wave speed: $c_L^2 = k_s a^2/m$.
c_T	Long-wavelength transverse wave speed: $c_T^2 = (1-\alpha)\,k_s a^2/m$.

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